FIT2179 Data Visualisation
Course Notes

MARKS AND CHANNELS

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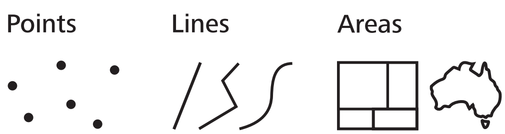
MARKS

Marks and channels are the building blocks for visualising table, network, field, and geometry datasets. Marks and channels together make invisible data attributes visible.

Marks are the fundamental geometric elements that are placed on a canvas to build data visualisations. There are five types of marks: point (zero-dimensional), line (one-dimensional), area (two-dimensional), volume (three-dimensional), and text.



Three-dimensional volume marks are mainly used for three-dimensional visualisation. Examples include cubes, spheres, or surfaces. This text focuses on two-dimensional visualisation and does not further discuss three-dimensional volume marks.

Text is also a fundamental graphical element that conveys information. However, whether text should be considered as a type of mark is debatable because it can be argued that the outlines of a text are areas.

CHANNELS

A channel (or visual variable) determines the visual appearance of marks. For example, a colour channel may show line marks on a map with red to represent highways.

When the author of a data visualisation chooses a visualisation method, they also implicitly select the marks and channels of a visualisation. For example, when selecting a bar chart for a table dataset, we use line marks, which form the individual bars of the chart, and the size channel, which varies the length of the bars to encode attribute values.

The author of a visualisation can also explicitly vary a channel. In the bar chart example, we could show bars representing past values with blue and bars representing forecast values in light grey. In this example, we use the colour channel to show a categorical attribute (that is, past or forecast).

The combination of marks and channels encodes data: invisible attribute values are encoded as visual signs that form a data visualisation. Readers of a data visualisation

TERMINOLOGY

The term *channel* is common in the data visualisation literature, but there are many synonyms consisting of combinations of visual/graphical/perceptual with channel/variable. A widely used synonym is *visual variable*, which is used in the cartographic literature.

invert the encoding: they observe the visual signs and gain an understanding of the data.

Not all channels are equally effective at encoding data attributes. Some channels are more effective than others.

There exist two major groups of channels: channels for ordered attributes, and channels for categorical attributes. Most channels can only be used for encoding either ordered attributes or categorical attributes.

There are six important channels: three distinct colour channels (hue, saturation, and luminance), the size channel (with distinctions for point, line, and area marks), the shape channel (mainly for point and line marks), and the position channel. There are additional channels, which are occasionally used for encoding data in a visualisation.

COLOUR HUE

Hue is the property that distinguishes the colours in a rainbow.

A rainbow transitions between these hues: red, orange, yellow, green, cyan, blue, violet.

Brown and pink are two examples of hues that are not visible on a rainbow. Brown is orange with reduced luminance, and pink is red with reduced saturation.

The colour gradient below gradually varies hue; all colours are fully saturated and luminance is constant.

The hue channel can encode categorical attributes but not ordered attributes, because there is no intuitive order in hues. For example, when comparing a blue and a green dot, we do not intuitively infer that the blue dot is encoding a higher value and the green dot a lower value (or vice versa).

Black and white are not colour hues. Black is the absence of light and white is the presence of light with all visible wavelengths.

COLOUR SATURATION

Saturation indicates how far away a colour is from a shade of grey.

A rainbow shows fully saturated colours.

Saturation varies between grey and a pure colour as a rainbow shows it. In the example below, hue and brightness do not change; only the saturation channel decreases from left to right. Minimum saturation is on the right side, where the blue colour has transitioned into grey.

COLOUR LUMINANCE, VALUE, BRIGHTNESS OR LIGHTNESS

Luminance/value/brightness/lightness refers to how bright or dark a colour appears.

For the purpose of data visualisation, colour luminance can also be called colour value, colour brightness or colour lightness. All terms express how dark or bright a colour is. In the example below, the hue is purple and colour luminance varies between dark and bright. Black on the left side is the minimum brightness, and white on the right side is the maximum brightness.

SIZE ▪■■

The size channel is for ordered attributes. We can vary the size of a dot (for instance, to distinguish small and large cities on a map), we can vary the width or length of a line (for instance, to show attribute values in a bar chart) , or we can shrink or enlarge an area (for instance, to encode values in a tree map).

There are three different size channels, one for each added dimension: length is a one-dimensional size, area is a two-dimensional size, and volume is a three-dimensional size.

SHAPE ▲ ★ ■

Marks with different shapes can indicate different categorical attributes.

POSITION

Spatial position is often overlooked when listing marks. The position channel can change the meaning of a mark. For example, a knob of a slider indicates a small or a large quantity, depending on its position. Another example is the position of a dot in a scatter-plot. Depending on where the dot is positioned, it encodes different attribute values.

OTHER CHANNELS

Besides the three colour channels, size, shape, and position, there are various other visual channels that can be applied to marks for visualising data.

Orientation, angle, and tilt can be considered synonyms. It orients or rotates a mark to visualise an attribute. For example, an arrow can be oriented to indicate an increasing ↗ or a decreasing ↘ value.

Transparency or opacity changes how a mark is blended with its background. Transparency is an important tool for creating aesthetically pleasing visualisations, but it is rarely used to encode data.

Curvature bends a mark. It is rarely used for visualising ordered attributes. I)))

Text can be varied with typographic channels. Examples include type weight (**bold** vs. thin) to show ordered attributes or style (*slanted* vs. upright) to show categorical attributes. These typographic channels are specific to text.

Various authors have extended the framework of visual marks and channels to include animation, sound, and tactile channels. However, animation and sound are rarely used for encoding data.

CHANNELS APPLIED TO MARKS

The following figure illustrates how channels and marks are combined. Note that the combinations of areas with size and shape channels result in cartograms, which are a group of specialised visualisation methods for geographic maps.

> **TERMINOLOGY**
>
> Note that the term *area* has two distinct meanings. On the one hand, an area is a type of mark that has a two-dimensional extent (such as a square or a region on a map). On the other hand, area is also a channel when referring to a two-dimensional size.

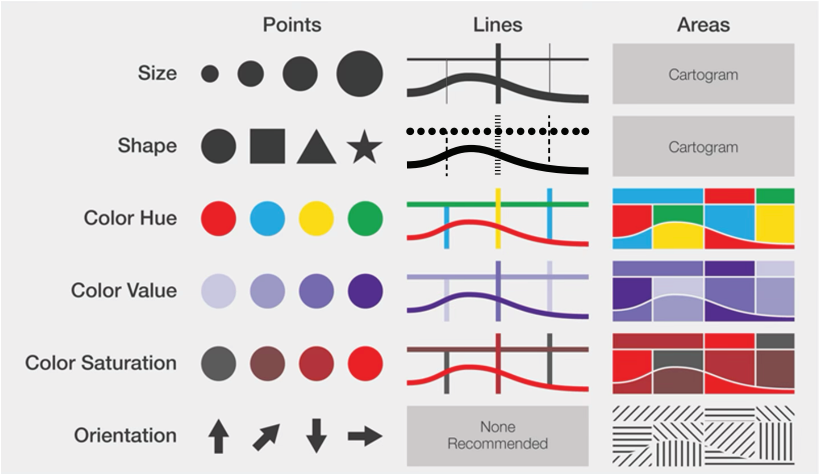


Image after: GIS&T Body of Knowledge www.ucgis.org/site/gis-t-body-of-knowledge

We can also apply more than one channel to a mark. For example, we can vary the size and the colour of a dot. The two channels could be redundant and encode the same data attribute, or they could be complementary and encode two different attributes.

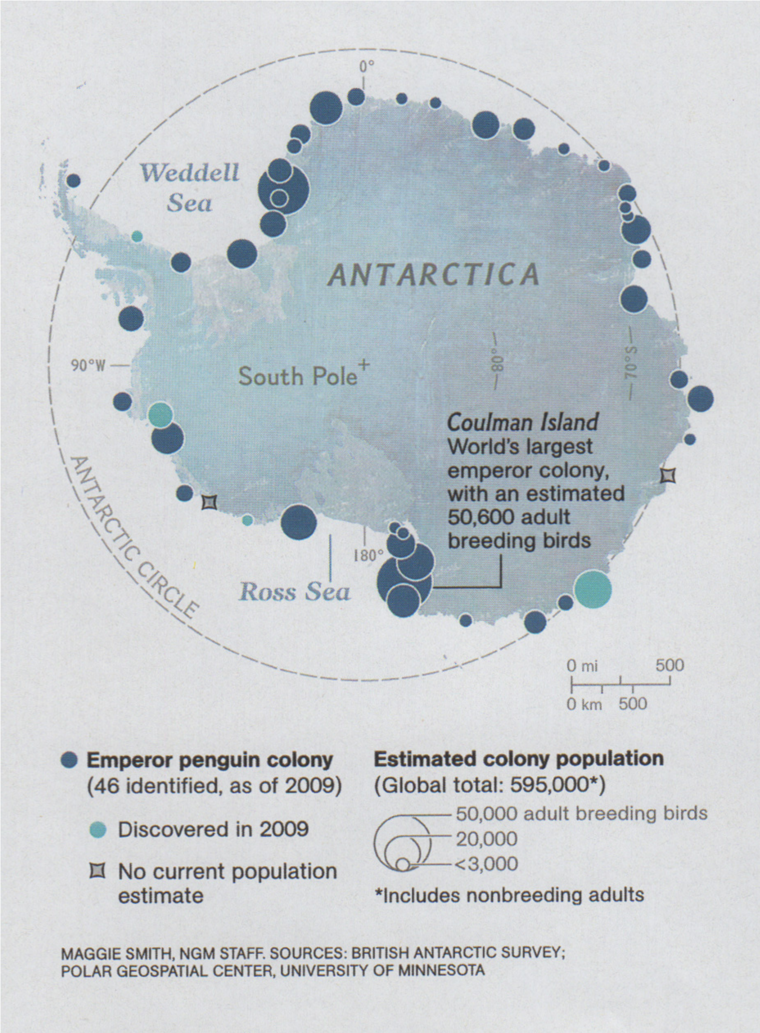
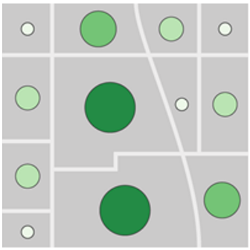


Image: National Geographic Magazine, November 2012.

When two channels are applied to a mark, we speak of bivariate encoding. Multivariate encoding is the term for two or more channels expressing different attributes.

The small map below shows an example with a single ordered attribute. The size of the circles, the brightness of the circles, and also the hue of the circles (yellow-green for small values and green for large values) vary to show a single attribute with a redundant encoding.



The large map showing penguin colonies also uses bivariate encoding, but varies the colour of the circles depending on a second attribute, which indicates when the colony was discovered.

CHANNELS RANKING

Some channels can only encode ordered attributes, and other channels can only encode categorical attributes. The following figure shows the two groups. The left column lists the most important channels for encoding ordered attributes. The right column shows the channels that can be used to visualize categorical attributes.

The ranking of the channels from most effective at the top to the least effective at the bottom is the result of many human subject studies. Cleveland and McGill published a foundational study that introduced this order by effectiveness.

Effective in this context means how accurately humans decode the channels.

The order from most effective to least effective is particularly important for ordered attributes. Position is the most effective channel for ordered attributes. That is, with position, humans are very good at decoding attribute values from a mark. If you look at the dots on the horizontal scales on the top-left, it is intuitive to understand why, as it is easy to estimate that the first dot is posi-

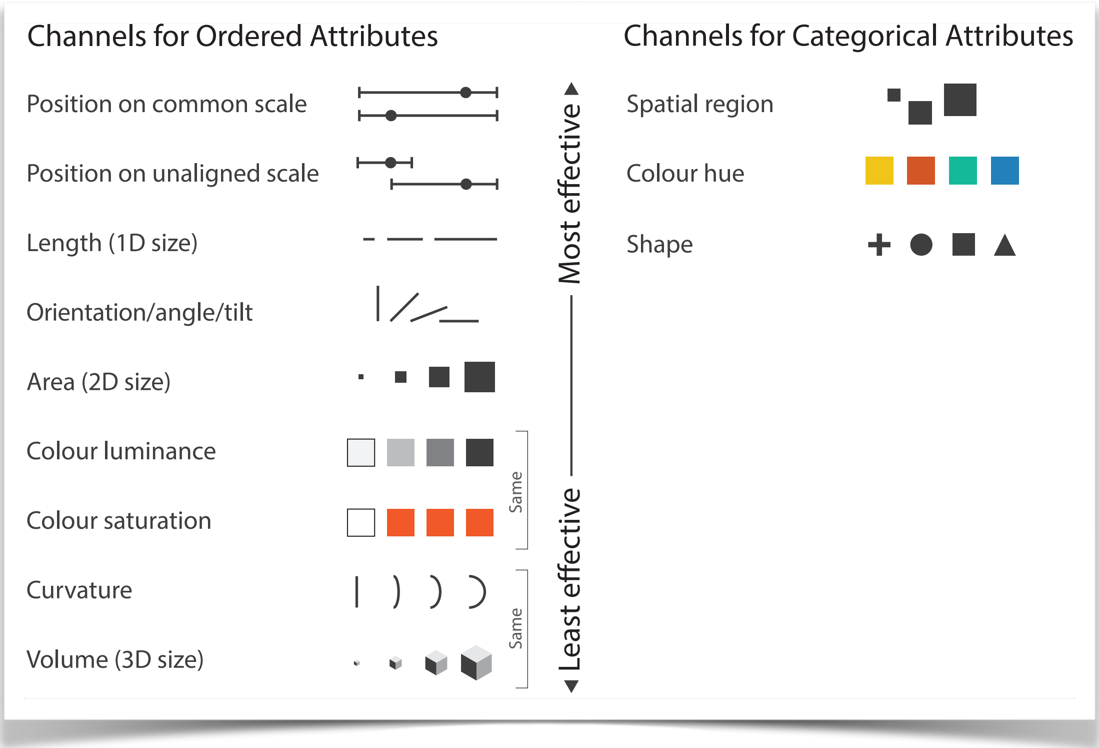


Image after: *Visualization Analysis and Design,* 2014 by Tamara Munzner.

tioned at about 80% of the total length, and the second value at about 20%.

If the reference scales are of different lengths or do not start at the same position, then estimating and reading positions is more difficult. That is, the decoding process is slightly less accurate.

There is a clear hierarchy among the three size channels. The length (or one-dimensional size) of lines is an effective channel. It is easy for humans to compare the length of horizontal or vertical bars. The area (or two-dimensional size) of a mark is clearly less accurate to interpret: Humans tend to overestimate the size of larger marks when comparing them to smaller marks. The lowest-ranked channel in the list for ordered data is volume (or three-dimensional size). Humans are particularly bad at accurately comparing volumes, even with simple shapes such as cubes or spheres. In summary, humans are very good at comparing the length of lines, less good at comparing areas, and not good at all at comparing volumes.

Of the three colour channels, only luminance and saturation can encode ordered attributes. (There is no intuitive order in colour hues.) Luminance is positioned before saturation, so luminance is more effective at showing ordered attributes. But luminance and saturation are both ranked quite low in this list.

The right column lists the channel appropriate for categorical attributes.

Spatial region is related to position, but applies to categorical attributes. An example is a Venn diagram. If a dot is inside a first circle, it belongs to the first category, and if the dot is inside the second circle, it belongs to the second category.

Colour hue and shape are both effective channels for showing different categories.

STEVENS' POWER LAW

A rationale for the ranking of channels is provided by Stevens' power law. Stevens' empirical law expresses the relationship between a physical stimulus and the perceived sensation created by the stimulus. The stimulus is a physical intensity, such as the area of circles in mm^2, and the perceived sensation is the area estimated by an average person.

The line chart illustrates that the relation between stimulus I and perceived sensation S is not linear for every channel. For example, length is close to linear ($S = I^1$), which means that humans can accurately judge the length of a mark. The length channel is therefore ranked highly. Area ($S = I^{0.7}$) is typically underestimated, which means that humans are less accurate at reading and comparing marks of different areas. They tend to underestimate the area of larger marks, and the area channel is accordingly ranked lower.

Colour brightness tends to be strongly underestimated *($S = I^{0.5}$)* and colour saturation strongly overestimated *($S = I^{1.7}$)*, which results in a low ranking for these channels

Note that Stevens' law ignores individual differences in how people match a stimulus to a sensation; there are generally large individual differences in this relationship.

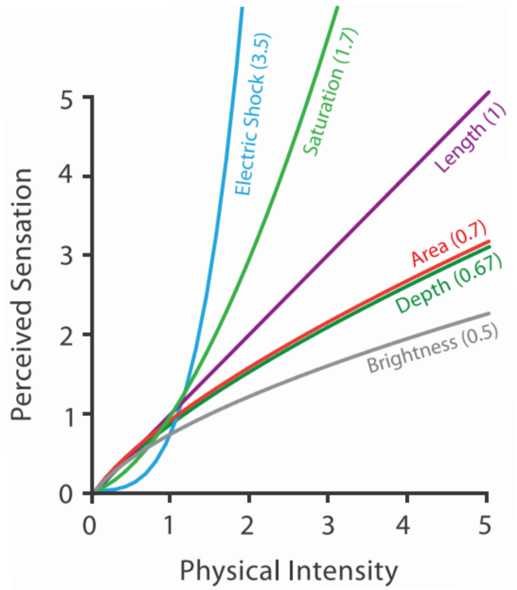


Image source: *Visualization Analysis and Design,* 2014 by Tamara Munzner.

FURTHER READING

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